IPC, Error and Event Contract Guidance

1. Rules

Adapt to the existing frozen SDK. Do not create duplicate contracts. Use explicit allowlisted methods; validate in main process; return typed results; never expose raw `ipcRenderer`; sanitize errors; use run IDs for progress/cancel.

2. Illustrative API surface

- `quote.create(request)`
- `quote.start(quoteId, insurerIds)`
- `quote.cancel(runId)`
- `quote.get(runId)`
- `quote.onProgress(callback)`
- `quote.addManualResult(input)`
- `quote.generatePdf(quoteId, options)`
- `history.list(filters)`
- `history.delete(quoteId)`
- `settings.get()`
- `settings.update(input)`
- `diagnostics.export()` (sanitized and user-reviewed)

Names are illustrative; existing code contracts take precedence.

3. Result envelope

```
type AppError = {
code: string;
message: string; // safe user-facing message
retryable: boolean;
correlationId?: string;
details?: Record<string, unknown>; // sanitized only
};

type AppResult<T> =
| { ok: true; data: T }
| { ok: false; error: AppError };
```

4. Progress event

`runId`, `insurerId`, `status`, optional `progressStep`, `safeMessage`, `occurredAt`. Never broadcast credentials, full member objects, session tokens or raw screenshots.

5. Error map

Code	Meaning	UI action
VALIDATION_FAILED	Input invalid	Correct highlighted fields
AUTH_REQUIRED	Login required	Sign in
OTP_REQUIRED	OTP required	Complete challenge
MANUAL_ACTION_REQUIRED	Human action needed	Follow permitted flow
PORTAL_UNAVAILABLE	Portal unreachable	Retry later/manual

Code	Meaning	UI action
NAVIGATION_TIMEOUT	Page timeout	Review and safe retry once
FIELD_NOT_FOUND	Expected field absent	Manual fallback/report
PARSE_FAILED	Ambiguous result	Do not rank; review
QUOTE_NOT_AVAILABLE	No quote returned	Review eligibility/details
RATE_LIMITED	Portal limiting requests	Stop and respect cooldown
CANCELLED	User cancelled	Start a new run if needed
STORAGE_ERROR	Local persistence failed	Protect data; stop risky writes
SECRET_STORE_UNAVAILABLE	Secure storage unavailable	Never save plaintext
PDF_GENERATION_FAILED	PDF failed	Review data/template and retry
INTERNAL_ERROR	Unexpected error	Share sanitized diagnostics

6. Cancellation/retry

Cancellation should be idempotent. Retry only classified transient failures and only where duplicate side effects are not possible. Preserve already validated results.